

Pulse Blow Unit

RoHS

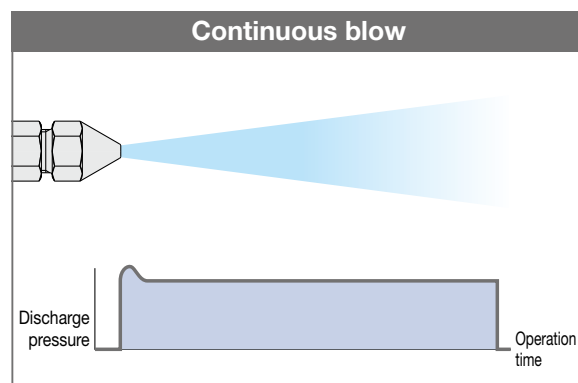
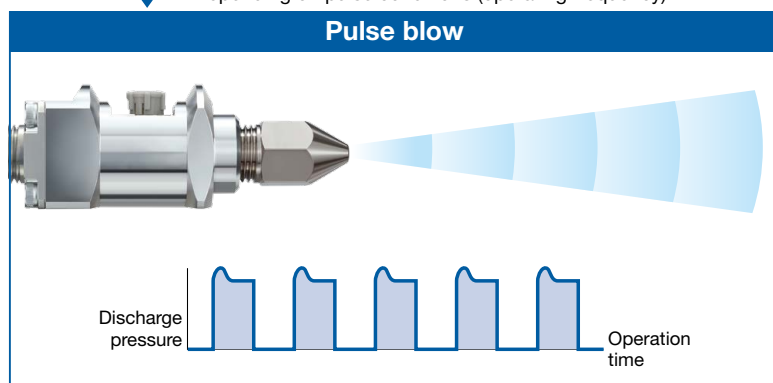
*Intermittent pulse blow improves removal efficiency by **35%** compared with continuous blow.*¹*

(*1 Based on the test by SMC.)

CO₂ emissions (Air consumption)

50% reduction

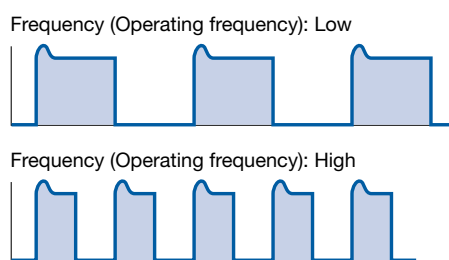
* Depending on pulse conditions (operating frequency)



Possible to adjust operating frequency

Frequency characteristics in relation to the number of needle rotations **p.1**

Push-lock type
Frequency adjustment knob (8 rotations)



External power supply is unnecessary.

A pulse blow is possible by the air supply.

Reduced pressure loss

The large effective area (10.3 mm²) relative to the recommended nozzle enables air blow with low pressure loss (Converted C value: 2.06 dm³/(s·bar))

Possible to use it with its bracket mounted



Nozzle for blowing

Combined use with a blow gun



Blow gun

Compact, Lightweight

Weight: Max. **97%** reduction

PU: **47 g** ← AXTS040: 1400 g

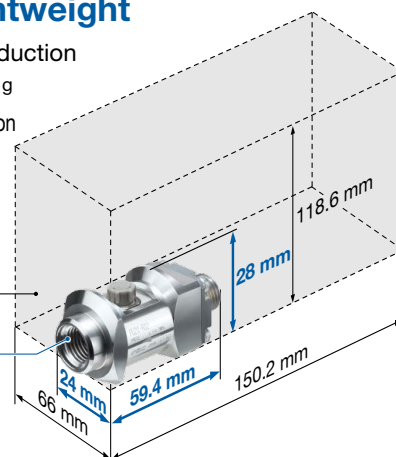
Volume: Max. **86%** reduction



1400 g

Pulse blow valve
AXTS040

47 g PU



Refer to the **Web Catalog** for details on the **KN series** nozzles for blowing and the **VMG series** blow gun.

PU Series

SMC

CAT.ES20-326B

Pulse Blow Unit

PU Series

RoHS



Specifications

Fluid	Air
Operating pressure range	0.3 to 0.7 MPa
Proof pressure	1.05 MPa
Ambient and operating fluid temperatures	-5 to 60°C (No freezing)
Frequency adjustment range	5 to 10 Hz
Port size	IN side port: R1/4 (male thread) OUT side port: Rc1/4 (female thread)
Weight	47 g
Body material	Aluminum
Recommended mounting nozzle dia. [mm]	ø2.0 or less
Effective area [mm ²]	10.3

How to Order

PU20-R02-B

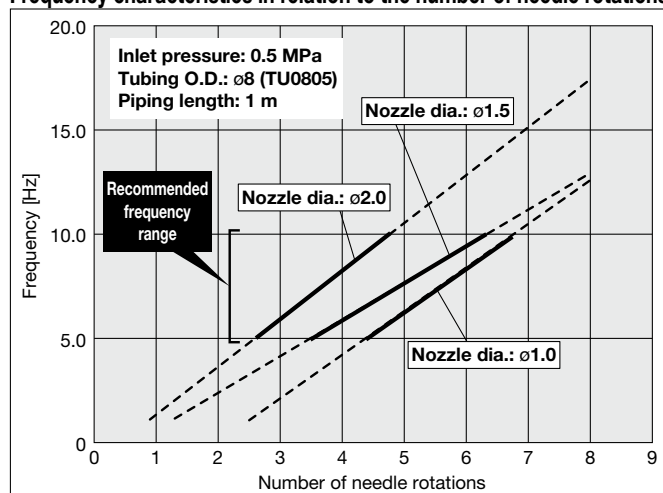
• Option

Nil	None
B	With bracket

• Connection thread (IN side)

R02	R1/4
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Frequency characteristics in relation to the number of needle rotations



Bracket Part Number (To Be Ordered Separately)

PU20P-B (Material: Resin)

Set Part Number for Pulse Blow Unit and Blow Gun

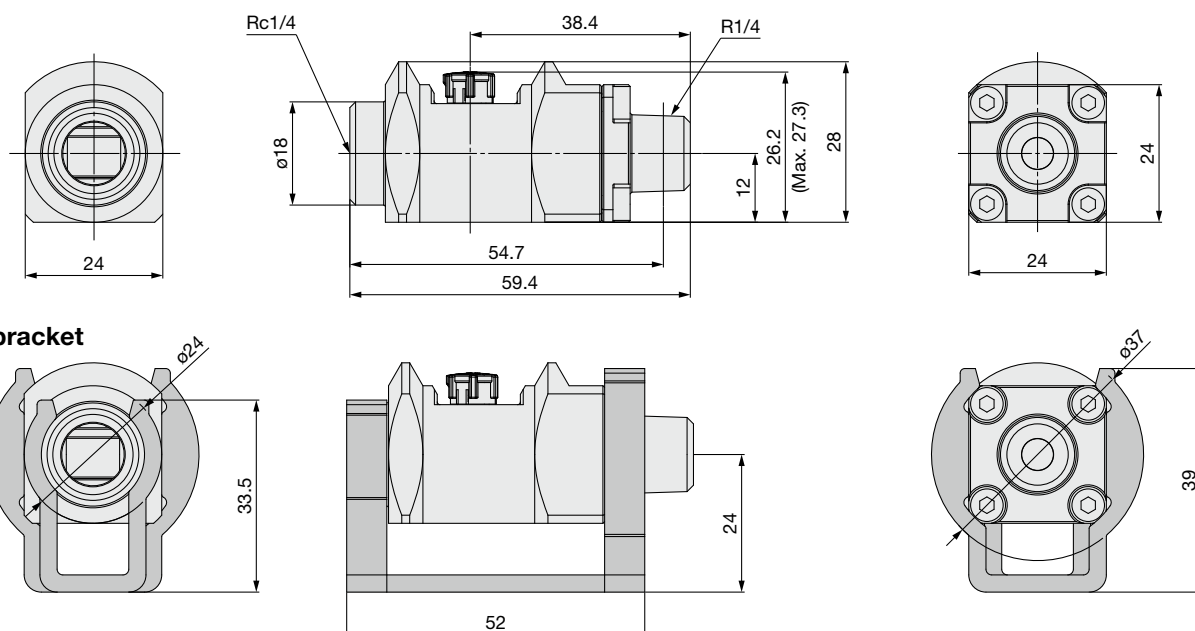
VMG standard model number - P



• With pulse blow unit

* The applicable nozzle effective area ranges from ø1 to ø2.
In addition, note that the extension nozzle is not applicable, as it prevents the product from operating properly. Refer to the [VMG series](#) blow gun catalog for details on applicable nozzles.

Dimensions





Specific Product Precautions

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For fittings and tubing precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Caution on Design

Warning

1. Confirm the specifications.

Products represented in this catalog are designed only for use in compressed air systems. Do not operate at pressures, temperatures, etc., beyond the range of specifications, as this can cause damage or malfunction.

2. Do not disassemble the product or make any modifications, including additional machining.

Doing so may cause human injury and/or an accident.

3. The product's frequency characteristics are representative values.

Be careful that the frequency varies according to individual differences. In addition, note that the recommended frequency range is 5 to 10 Hz.

4. Be careful of the operating pressure range.

This product's operating pressure range is from 0.3 to 0.7 MPa. Check the operating environment prior to use.

Caution

1. The recommended nozzle diameter for the product is $\phi 1$ to $\phi 2$.

It is recommended to screw the nozzle directly into the product's secondary side.

The recommended pipe tube inner diameter on the supply side is 5 mm or more and the recommended pipe tube length is 5 m or less.

When using tubes with an inner tube diameter of less than 5 mm, use nozzles with a nozzle diameter of $\phi 1$.

Normal pulse operation may not be possible when using piping conditions other than those recommended.

Please refer to the blowguns catalogue (VMG series) for compatible nozzles

2. Do not use the product with flammable, explosive, or toxic substances such as gas, gas fuel, and refrigerants.

Such substances may permeate from inside the product.

3. Operation in a low temperature condition

Take appropriate measures to avoid freezing of drainage, moisture etc. in low temperature.

If drainage or moisture solidifies or freezes, a malfunction may result.

Caution

4. Do not supply air pressure from other ports than supply-side port.

Malfunction may occur when the pressure is supplied from the wrong ports.

5. Lubricant inside the product

Grease is applied to the inner parts of the product.

Grease may enter the downstream side of the product.

6. The duty ratio of the product is approximately 50%.

Please note, this may vary depending on the operating conditions.

Within the product specification range, increasing the nozzle diameter and lowering the frequency tends to improve the product's air-saving performance.

7. The product operates based on back pressure.

If piping is connected to the secondary side of the product, or if there is volume on the secondary side, the product may not operate properly. When using the product, it is recommended that the nozzle be screwed directly into the product.

In addition, note that even if the nozzle diameter is $\phi 1$ to $\phi 2$, if multiple outlets cause the total effective area to exceed $\phi 2$, the product may not operate properly.

If a restriction is installed on the secondary side and piping is connected, be sure to verify that the product operates properly before use.



Safety Instructions

These safety instructions are intended to prevent hazardous situations and/or equipment damage. These instructions indicate the level of potential hazard with the labels of “**Caution**,” “**Warning**” or “**Danger**.” They are all important notes for safety and must be followed in addition to International Standards (ISO/IEC)*1), and other safety regulations.

 **Danger :** **Danger** indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.

 **Warning:** **Warning** indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.

 **Caution:** **Caution** indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.

*1) ISO 4414: Pneumatic fluid power - General rules and safety requirements for systems and their components
ISO 4413: Hydraulic fluid power - General rules and safety requirements for systems and their components
IEC 60204-1: Safety of machinery - Electrical equipment of machines - Part 1: General requirements
ISO 10218-1: Robots and robotic devices - Safety requirements for industrial robots - Part 1: Robots etc.

Warning

1. The compatibility of the product is the responsibility of the person who designs the equipment or decides its specifications.

Since the product specified here is used under various operating conditions, its compatibility with specific equipment must be decided by the person who designs the equipment or decides its specifications based on necessary analysis and test results. The expected performance and safety assurance of the equipment will be the responsibility of the person who has determined its compatibility with the product. This person should also continuously review all specifications of the product referring to its latest catalog information, with a view to giving due consideration to any possibility of equipment failure when configuring the equipment.

2. Only personnel with appropriate training should operate machinery and equipment.

The product specified here may become unsafe if handled incorrectly. The assembly, operation and maintenance of machines or equipment including our products must be performed by an operator who is appropriately trained and experienced.

3. Do not service or attempt to remove product and machinery/equipment until safety is confirmed.

1. The inspection and maintenance of machinery/equipment should only be performed after measures to prevent falling or runaway of the driven objects have been confirmed.
2. When the product is to be removed, confirm that the safety measures as mentioned above are implemented and the power from any appropriate source is cut, and read and understand the specific product precautions of all relevant products carefully.
3. Before machinery/equipment is restarted, take measures to prevent unexpected operation and malfunction.

4. SMC products cannot be used beyond their specifications. They are not developed, designed, and manufactured to be used under the following conditions or environments. Use under such conditions or environments is not allowed.

1. Conditions and environments outside of the given specifications, or use outdoors or in a place exposed to direct sunlight.
2. Use for nuclear power, railways, aviation, space equipment, ships, vehicles, military application, equipment affecting human life, body, and property, combustion equipment, entertainment equipment, emergency shut-off circuits, press clutches, brake circuits, safety equipment, etc., and use for applications that do not conform to standard specifications such as catalogs and operation manuals.
3. Use for interlock circuits, except for use with double interlock such as installing a mechanical protection function in case of failure. Please periodically inspect the product to confirm that the product is operating properly.

Caution

SMC develops, designs, and manufactures products to be used for automatic control equipment, and provides them for peaceful use in manufacturing industries.

Use in non-manufacturing industries is not allowed.

Products SMC manufactures and sells cannot be used for the purpose of transactions or certification specified in the Measurement Act of each country. The new Measurement Act prohibits use of any unit other than SI units in Japan.

Limited warranty and Disclaimer/ Compliance Requirements

The product used is subject to the following “Limited warranty and Disclaimer” and “Compliance Requirements”.

Read and accept them before using the product.

Limited warranty and Disclaimer

1. The warranty period of the product is 1 year in service or 1.5 years after the product is delivered, whichever is first.*2)
Also, the product may have specified durability, running distance or replacement parts. Please consult your nearest sales branch.
2. For any failure or damage reported within the warranty period which is clearly our responsibility, a replacement product or necessary parts will be provided.
This limited warranty applies only to our product independently, and not to any other damage incurred due to the failure of the product.
3. Prior to using SMC products, please read and understand the warranty terms and disclaimers noted in the specified catalog for the particular products.

*2) **Suction cups (Vacuum pads) are excluded from this 1 year warranty.**

A suction cup (vacuum pad) is a consumable part, so it is warranted for a year after it is delivered.

Also, even within the warranty period, the wear of a product due to the use of the suction cup (vacuum pad) or failure due to the deterioration of rubber material are not allowed by the limited warranty.

Compliance Requirements

1. The use of SMC products with production equipment for the manufacture of weapons of mass destruction (WMD) or any other weapon is strictly prohibited.
2. The exports of SMC products or technology from one country to another are governed by the relevant security laws and regulations of the countries involved in the transaction. Prior to the shipment of a SMC product to another country, assure that all local rules governing that export are known and followed.

Revision History

Edition B * Added tube conditions for the frequency characteristics in relation to the number of needle rotations.
* Added air blow conditions for the reduction of pressure loss.
* Added a recommended frequency range for the Specific Product Precautions.

 **Safety Instructions** Be sure to read the “Handling Precautions for SMC Products” (M-E03-3) and “Operation Manual” before use.